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# React `useEffect` Hooks

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The `useEffect` Hook allows you to perform side effects in your components.

Some examples of side effects are: fetching data, directly updating the DOM, and timers.

`useEffect` accepts two arguments. The second argument is optional.

```
useEffect(<function>, <dependency>)
```

Let's use a timer as an example.

## Example:

Get your own React.js Server

Use `setTimeout()` to count 1 second after initial render:

```
import { useState, useEffect } from 'react';
import { createRoot } from 'react-dom/client';
```

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```
    setTimeout(() => {
      setCount((count) => count + 1);
    }, 1000);
  });

  return <h1>I've rendered {count} times!</h1>;
}
```

Run Example »

But wait!! It keeps counting even though it should only count once!

`useEffect` runs on every render. That means that when the count changes, a render happens, which then triggers another effect.

This is not what we want. There are several ways to control when side effects run.

We should always include the second parameter which accepts an array. We can optionally pass dependencies to `useEffect` in this array.

## Example

1. No dependency passed:

```
useEffect(() => {
  //Runs on every render
});
```

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```
useEffect(() => {
  //Runs only on the first render
}, []);
```

## Example

3. Props or state values:

```
useEffect(() => {
  //Runs on the first render
  //And any time any dependency value changes
}, [prop, state]);
```

So, to fix this issue, let's only run this effect on the initial render.

## Example:

Only run the effect on the initial render:

```
import { useState, useEffect } from 'react';
import { createRoot } from 'react-dom/client';

function Timer() {
  const [count, setCount] = useState(0);

  useEffect(() => {
    setTimeout(() => {
      setCount((count) => count + 1);
    }, 1000);
```

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```
createRoot(document.getElementById('root')).render(
  <Timer />
);
```

Run Example »

## Example:

Here is an example of a `useEffect` Hook that is dependent on a variable. If the `count` variable updates, the effect will run again:

```
import { useState, useEffect } from 'react';
import { createRoot } from 'react-dom/client';

function Counter() {
  const [count, setCount] = useState(0);
  const [calculation, setCalculation] = useState(0);

  useEffect(() => {
    setCalculation(() => count * 2);
  }, [count]); // <- add the count variable here

  return (
    <>
      <p>Count: {count}</p>
      <button onClick={() => setCount((c) => c + 1)}>+</button>
      <p>Calculation: {calculation}</p>
    </>
  );
}

createRoot(document.getElementById('root')).render(
```

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If there are multiple dependencies, they should be included in the `useEffect` dependency array.

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# Effect Cleanup

SHARE

Some effects require cleanup to reduce memory leaks.

Timeouts, subscriptions, event listeners, and other effects that are no longer needed should be disposed.

We do this by including a return function at the end of the `useEffect` Hook.

## Example:

Clean up the timer at the end of the `useEffect` Hook:

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```
function Timer() {
  const [count, setCount] = useState(0);

  useEffect(() => {
    let timer = setTimeout(() => {
      setCount((count) => count + 1);
    }, 1000);

    return () => clearTimeout(timer)
  }, []);

  return <h1>I've rendered {count} times!</h1>;
}

createRoot(document.getElementById('root')).render(
  <Timer />
);
```

Run Example »

**Note:** To clear the timer, we had to name it.

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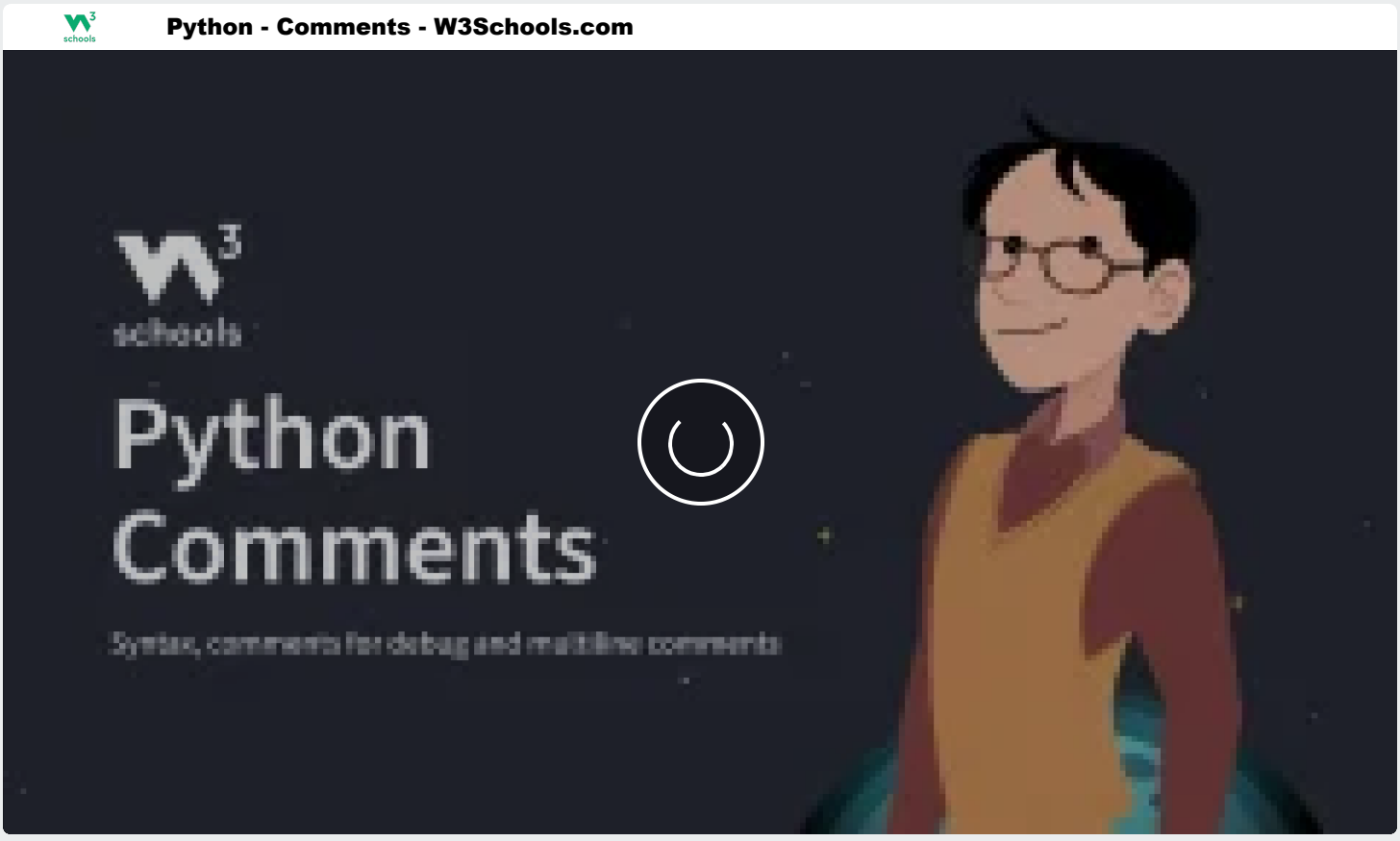
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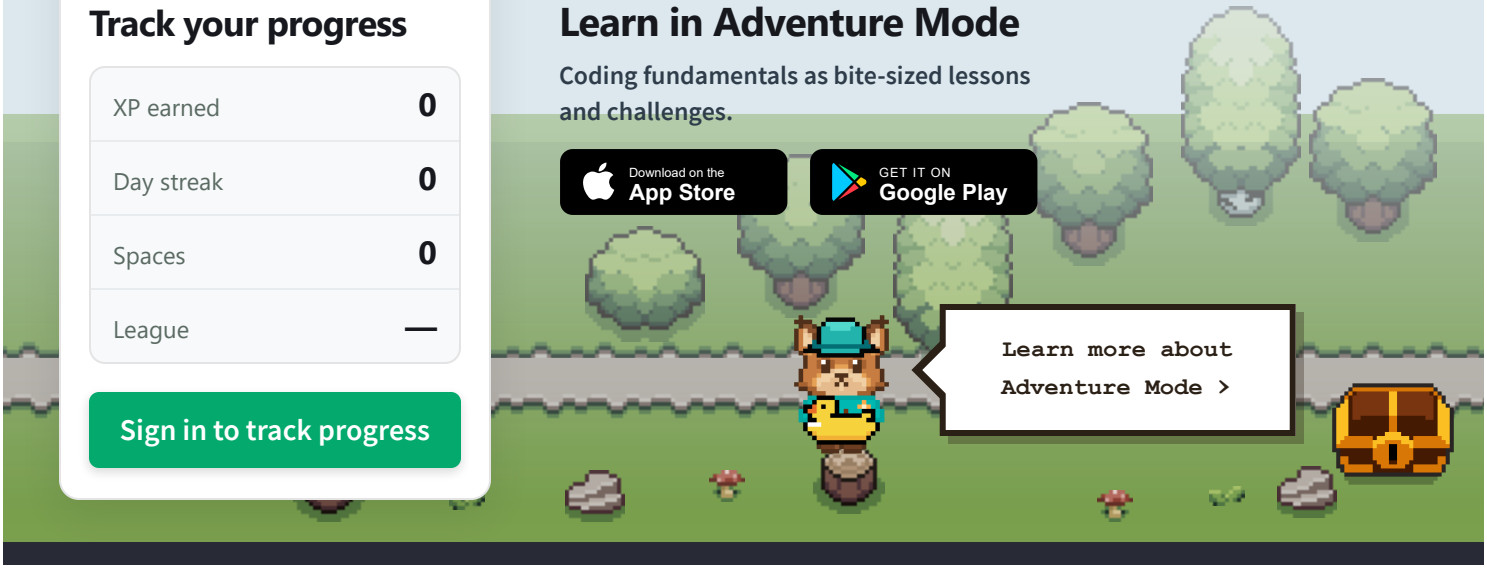
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